

Problem Set 2

(BUSFIN 8200: Asset Pricing Foundations)

Problem Set Rules:

- Your problem set solution must be delivered in the form of a typed report in pdf (not handwritten).
- You must also provide a Git repository with all your files and results. It includes the source files used to generate the pdf (e.g., the .tex script) as well as all coding scripts used to produce the results (e.g., the .R or .py scripts). I need to be able to reproduce your results and pdf from the source files in your Git repository.
- You are allowed to consult any material while working on the problem set (except solutions created by individuals not currently taking the course).
- You are allowed to talk with anyone while working on your problem set. In particular, I highly recommend that you talk with your classmates about the problem set. Research requires collaboration. It is fine if you exchange information and even jointly produce parts of the problem set codes/derivations. However, you have to submit your own solutions.
- Do not abuse the rule above. While you can work together to produce the problem set solution (or even use the solution of a classmate as an initial point in developing your own solution), you cannot simply copy a question solution from a classmate.
- You can also use AI when working on your problem set. However, you need to follow the rules in the “BUSFIN 8200 Problem Sets: AI Policy” document.

Datasets

(BM Dataset for Questions 2, 3, and 4) Download the returns (from July 1927 to the latest date) on the book-to-market decile portfolios from the Ken French data library (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) under columns “Lo 10” to “Hi 10” from the .csv file that can be downloaded in the link under “Portfolios Formed on Book-to-Market”. To calculate excess returns, use the risk-free proxy under the column “RF” from the file that can be downloaded in the link under “Fama/French 3 Factors”.

(MOM Dataset for Questions 2, 3, and 4) Download the returns (from July 1927 to the latest date) on the momentum decile portfolios from the Ken French data library (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) under columns “Lo PRIOR” to “Hi PRIOR” from the .csv file that can be downloaded in the link under “10 Portfolios Formed on Momentum”. To calculate excess returns, use the risk-free proxy under the column “RF” from the file that can be downloaded in the link under “Fama/French 3 Factors”.

(EQ Dataset for Question 3) The file “EQ Dataset.csv” has the following columns:

- YEAR: the year associated with the observation
- MONTH: the month associated with the observation
- dp: the log of the dividend-to-price ratio ($dp_t = \log(D_t/P_t)$), with D_t reflecting the sum of the dividends paid by the total US equity market (i.e., firms available on CRSP) over the 12 months ending in month t . Dividends account for M&A paid in cash following the procedure described in Gonçalves (2021)
- dg: the log of the annual dividend growth ($\Delta d_t = \log(D_t/D_{t-12})$), with D_t reflecting the sum of the dividends paid by the total US equity market (i.e., firms available on CRSP) over the 12 months ending in month t . Dividends account for M&A paid in cash following the procedure described in Gonçalves (2021)
- rf: the log of the annual (deflated) risk-free return ($r_{f,t} = \log(R_{f,t})$) ending in the given month
- re: the log of the annual (deflated) equity market return ($r_{e,t} = \log(R_{e,t})$) ending in the given month

Note: the dataset contains monthly observations of annual variables. So, the empirical analyses in Question 3 has to be done accordingly.

1 Question 1

Consider an economy with three state of nature and three financial assets:

Table 1
States of Nature and Financial Assets in our Economy

States of Nature	$n = 1$	$n = 2$	$n = 3$	
Probabilities	$\pi(1) = 1/4$	$\pi(2) = 1/2$	$\pi(3) = 1/4$	
Financial Asset	Payoff under Each State of Nature			Price
A	10	20	30	16
B	10	10	10	8
C	0	10	10	6

- 1a) Do we have Complete Markets in this economy?¹
- 1b) Suppose that the Law of One Price (LOP) holds in this economy. Calculate the price of each Contingent Claim (or Arrow-Debreu security) in this economy.²
- 1c) Suppose that the LOP holds in this economy and introduce a new asset (referred to as asset D) with payoff vector $\text{Payoff}(D) = (30, 20, 10)$ for our three states of nature $n = (1, 2, 3)$. Calculate the price of asset D in this economy
- 1d) Calculate the Stochastic Discount Factor (SDF) realization under each of the three states of nature in this economy. Does this economy satisfy No-Arbitrage (explain)? Is there anything interesting about the SDF in this economy (if so, explain what it means in economy terms)?
- 1e) Show that the price that the SDF assigns to asset D in this economy is the same as the price that you calculated in Question (1c)

¹Note that we have Complete Markets if and only if for each state of nature n we can find a portfolio (i.e., linear combination of the original assets) that provides a payoffs of 1 in that state of nature and 0 in all other states of nature. Letting $\mathbf{X}$ reflect the matrix of $N = 3$ non-redundant asset payoffs, we can state this condition in linear algebra terms: we have Complete Markets if and only if we can find a unique matrix Ω such that $\mathbf{X} \Omega = \mathbf{I}$.

²Similar to Question (1a), it is easier to solve Question (1b) if we use linear algebra notation. Specifically, let (i) $\mathbf{X}$ reflect the matrix of $N = 3$ non-redundant asset payoffs, (ii) P reflect the price vector for the original assets, and (iii) P_c reflect the price vector for the contingent claims. You need to use the LOP to solve for P_c as a function of $\mathbf{X}$ and P .

2 Question 2

- 2a)** Find the SDF $M_t^* = a - b'xR_t = a - xR_t^*$ that unconditionally prices the excess returns of the 10 portfolios in the BM Dataset (ex-post). That is, find the SDF that satisfies $\hat{\mathbb{E}}[M^* \cdot xR] = 0$ for the xR vector in the BM Dataset.³ Provide a scatterplot with the portfolio deciles (1 to 10) in the x-axis and the respective b coefficients in the y-axis. Describe the pattern you observe in the scatterplot and explain why this pattern emerges.
- 2b)** Repeat Question (2a), but this time using the excess returns of the 10 portfolios in the MOM Dataset. Calculate and report the correlation between the M_t^* from this question and the M_t^* from Question (2a). Interpret this correlation in economic terms.
- 2c)** Repeat Question (2a), but this time using the excess returns of the 20 portfolios in the merge of the BM Dataset and the MOM Dataset. For the scatterplot, still have the portfolio deciles (1 to 10) in the x-axis, but use different colors for the dots associated with different sorting variables (blue for *BM* portfolios and red for *MOM* portfolios). Calculate and report the Sharpe ratios of the xR_t^* portfolios obtained in Questions (2a), (2b) and (2c). Explain what you observe and why it is (or it is not) in line with a key SDF property we learned about in the lectures.
- 2d)** For each portfolio p (covering the 20 portfolios from Question (2c)), calculate (and report in a table) the pricing errors $\hat{\alpha}_p = \hat{\mathbb{E}}[xR_p] - \hat{\text{Cov}}[-M^*, xR_p]$ under each M_t^* above (from Questions (2a), (2b), and (2c)). Describe the results and explain what we learn from them.
- 2e)** Repeat the M^* construction in Question (2c), but using data until December 1989. Then, apply the $\hat{a}$ and $\hat{b}$ obtained from this exercise to construct the out-of-sample SDF $M_{OS,t}^* = \hat{a} - \hat{b}'xR_t$ from January 1990 to the end of the sample. For each portfolio p (covering the 20 portfolios from Question (2c)), calculate (and report in a table) the pricing error $\hat{\alpha}_{OS,p} = \hat{\mathbb{E}}[xR_p] - \hat{\text{Cov}}[-M_{OS}^*, xR_p]$ using data from January 1990 to the end of the sample to estimate $\hat{\alpha}_{OS,p}$. Describe the results and explain what we learn from them.

³Let's use the normalization $\mathbb{E}[M^*] = 1$ (throughout Question 2). Note that we need a normalization for $\mathbb{E}[M^*]$ because if $\mathbb{E}[M^* \cdot xR] = 0$ then $\mathbb{E}[(\theta \cdot M^*) \cdot xR] = 0$ for any θ . Intuitively, since you are constructing an SDF that is only required to price excess returns, you are not using any information on the risk-free asset R_f (or more broadly the zero-beta asset R_0), and thus cannot identify $\mathbb{E}[M^*]$. If you also required the SDF to price a return (R_j whether $R_j = R_0$ or not), then you would also be able to identify $\mathbb{E}[M^*]$.

3 Question 3

Consider a representative investor with wealth W_t in a two-period economy (t and $t + 1$). At time t , the investor decides how to allocate his/her wealth across J assets (with return vector R_t) by solving the problem

$$\begin{aligned} \omega_t^* &= \underset{\omega_t}{\operatorname{argmax}} \mathbb{E}_t[U(W_{t+1})] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot R_{w,t+1} = W_t \cdot \omega_t' R_{t+1} \\ &\quad \mathbf{1}' \omega_t = 1 \end{aligned} \quad (3.1)$$

Letting, $R_{f,t}$ be a baseline asset (that does not need to be risk-free), we can equivalently write this problem as⁴

$$\varpi_t^* = \underset{\varpi_t}{\operatorname{argmax}} \mathbb{E}_t[U(W_{t+1})] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot R_{w,t+1} = W_t \cdot (R_{f,t+1} + \varpi_t' x R_{t+1}) \quad (3.2)$$

where xR_t is the vector of asset excess returns relative to $R_{f,t}$.

- 3a)** Using a power utility function, $U(W) = W^{1-\gamma}/(1-\gamma)$, with γ representing the constant relative risk aversion, show that the SDF $M_t = R_{w,t}^{-\gamma}$ prices the vector of excess returns (i.e., show that $\mathbb{E}_t[R_{w,t+1}^{-\gamma} \cdot xR_{t+1}] = \mathbf{0}$).
- 3b)** Estimate γ by GMM using the unconditional SDF equation $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$.⁵ Use the proxies $R_{w,t} = R_{e,t} = e^{r_{e,t}}$ and $R_{f,t} = e^{r_{f,t}}$, with r_e and r_f from the EQ Dataset. Report the γ estimate as well as its standard error. When estimating $a\text{Var}[\hat{\gamma}]$, estimate S using Newey and West (1987, 1994).
- 3c)** Estimate γ by GMM using the unconditional SDF equations $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\mathbb{E}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$. The second equation reflects the unconditional pricing of a managed portfolio with $z_t = \mathbb{E}_t^{OS}[xR_e]/\sigma_e^2$ as the trading instrument (and it captures a conditional implication of the model).⁶ Use an identity weighting matrix when minimizing pricing errors (so that the GMM estimate implicitly uses $\hat{A} = \hat{G}'$). Produce a table that reports the γ estimate as well as its standard error. The table should also report the value and t-statistic for the pricing error of each moment condition as well as the p-value for a χ^2 test with the null hypothesis that the pricing errors are jointly zero. When estimating $a\text{Var}[\hat{\gamma}]$ and $a\text{Var}[\hat{\mathbb{E}}[\epsilon(\hat{\gamma})]]$, estimate S using Newey and West (1987, 1994).

⁴Note that $\omega_t = (1 - \mathbf{1}' \varpi_t, \varpi_t)$, and thus $\mathbf{1}' \omega_t = 1$ holds regardless of the $\mathbf{1}' \varpi_t$ value. So, the formulation in Equation 3.2 is equivalent to the formulation in Equation 3.1, but it is easier to solve since we have one less constraint.

⁵Note that we have a single moment condition and a single parameter. However, it is possible that no value of γ is able to set $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, and thus you still need to use the GMM minimization problem $\hat{\gamma} = \operatorname{argmin} \hat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}]$. This minimization problem implies that the GMM estimate solves $\hat{G} \cdot \hat{\mathbb{E}}[R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t}] = 0$ where $\hat{G} = \hat{\mathbb{E}}[\partial_\gamma(R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t})]$.

⁶ σ_e^2 should be set to the unconditional variance of $R_{e,t}$ over the sample while $\mathbb{E}_t^{OS}[xR_e]$ is the out-of-sample forecast of $xR_{e,t+1}$ obtained from Question (2e) of Problem Set 1. In this case, we can interpret $z_{t-1} \cdot xR_{w,t}$ as the excess return on the maximum Sharpe ratio portfolio that can be formed with $R_{e,t}$ and $R_{f,t}$ at each time t using D_t/P_t as a trading signal (under the assumption that expected returns vary over time while volatility does not).

- 3d)** Repeat Question (3c), but this time using a two-step GMM in which the $\hat{S}^{-1}$ from Question (3c) is used as the weighting matrix when minimizing pricing errors (so that the feasible efficient GMM estimate implicitly uses $\hat{A} = \hat{G}'\hat{S}^{-1}$).
- 3e)** Repeat Question (3c), but this time using the moment condition vector $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_t] = \mathbf{0}$, where xR includes xR_w as well as the excess returns on the 20 decile portfolios in the merge of the BM Dataset and the MOM Dataset.

4 Question 4

- 4a) The SDF $M_{t+1} = R_{w,t+1}^{-\gamma}$ from Question 3 reflects a non-linear CAPM. Show that we also have (as an approximation) a conditionally linear CAPM, $M_{t+1} \approx a_t - b_t \cdot R_{w,t+1}$, and provide expressions for a_t and b_t . The advantage of a linear SDF is that it leads to a simpler GMM estimator. Highlight a disadvantage of the $M_{t+1} \approx a_t - b_t \cdot R_{w,t+1}$ relative to the original $M_{t+1} = R_{w,t+1}^{-\gamma}$ (beyond the fact that the linear SDF is an approximation).
- 4b) For simplicity, let's consider the SDF model $M_t = a - b \cdot R_{w,t}$ with the normalization $\mathbb{E}[M] = 1$, which can be written as $M_t = 1 - b \cdot (R_{w,t} - \mathbb{E}[R_w])$. Estimate b by GMM with the unconditional SDF equation $\mathbb{E}[M \cdot xR_t] = \mathbf{0}$, where xR includes xR_w as well as the excess returns on the 20 decile portfolios in the merge of the BM Dataset and the MOM Dataset. Use an identity weighting matrix when minimizing pricing errors (so that the GMM estimate implicitly uses $\hat{A} = \hat{G}'$). Produce a table that reports the b estimate as well as its standard error. The table should also report the value and t-statistic for the pricing error of each moment condition as well as the p-value for a χ^2 test with the null hypothesis that the pricing errors are jointly zero. When estimating $a\text{Var}[\hat{\gamma}]$ and $a\text{Var}[\hat{\mathbb{E}}[\epsilon(\hat{\gamma})]]$, estimate S using Newey and West (1987, 1994).
- 4c) Repeat Question (4b), but this time using the efficient GMM. Note that in this case you have an analytical expression for the GMM estimate of b and can easily compute it (without solving any numerical optimization problem).
- 4d) Estimate λ and α_j for each asset in Question (4b) using time series regressions for the linear beta pricing model implied by the $M_t = a - b \cdot R_{w,t}$ SDF. Produce a table that reports the λ estimate as well as its standard error (using Newey and West (1987, 1994) on a time series regression of $R_{w,t}$ on a constant). The table should also report the value and t-statistic for each α_j (using Newey and West (1987, 1994) on the respective time series factor regression). Finally, the table should also report the p-value for the GRS test with the null hypothesis that the pricing errors are jointly zero.
- 4e) Estimate λ and α_j for each asset in Question (4b) using Fama-MacBeth regressions for the linear beta pricing model implied by the $M_t = a - b \cdot R_{w,t}$ SDF. Produce a table that reports the λ estimate as well as its standard error. The table should also report the value and t-statistic for each α_j . The standard error for $\hat{\lambda}$ ($\hat{\alpha}_j$) should come from a time series regression of $\hat{\lambda}_t$ ($\hat{\alpha}_{j,t}$) on a constant with Newey and West (1987, 1994) standard errors.

References

- Gonçalves, A. S. (2021). “Reinvestment Risk and the Equity Term Structure”. In: *Journal of Finance* 76.5, pp. 2153–2197.
- Newey, W. K. and K. D. West (1987). “A Simple, Positive-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix”. In: *Econometrica* 55.3, pp. 703–708.
- Newey, W. K. and K. D. West (1994). “Automatic Lag Selection in Covariance Matrix Estimation”. In: *Review of Economic Studies* 61.4, pp. 631–653.